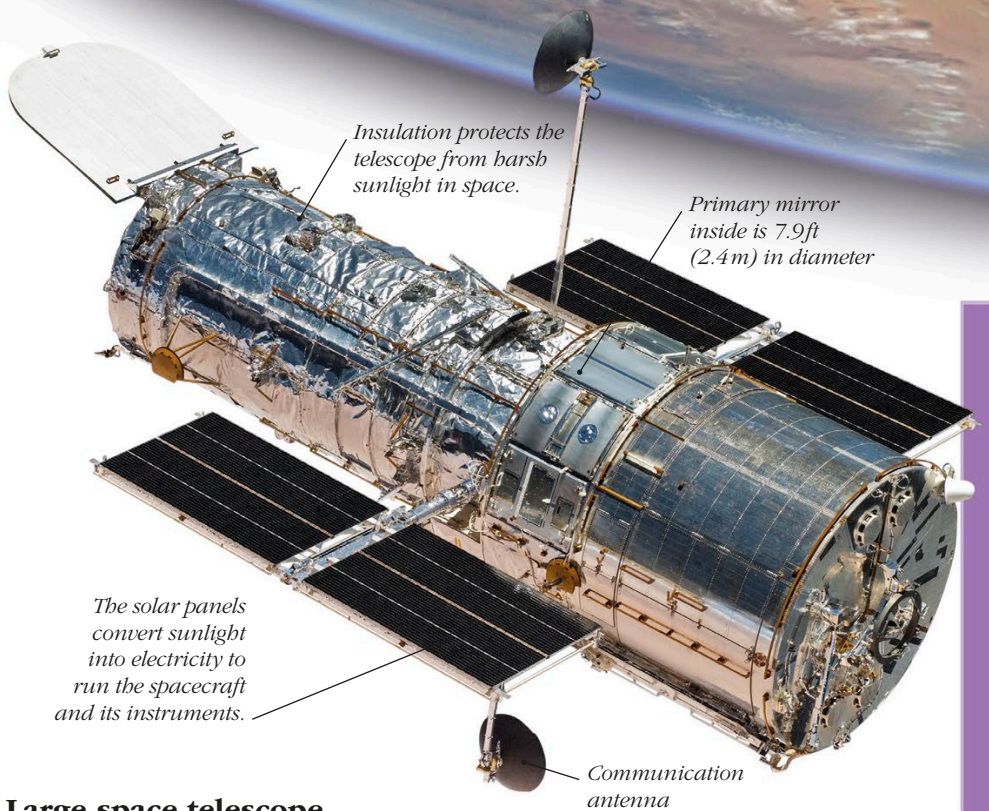


Observing from orbit

- **What?** Infrared Astronomical Satellite (IRAS)
- **Who?** NASA
- **Where and when?** Low Earth Orbit, 1983

From the late 1940s, rocket-borne radiation detectors revealed that space was filled with invisible rays that are absorbed by Earth's atmosphere. Ultraviolet satellite observatories launched since the 1960s study some of these rays, while infrared satellites, pioneered by IRAS, use specially cooled telescopes to detect weak heat radiation and map the night sky.



SPACE

Large space telescope

- **What?** Hubble Space Telescope
- **Who?** NASA
- **Where and when?** Low Earth Orbit, launched 1990

In 1946, the American astrophysicist Lyman Spitzer was the first to suggest putting a telescope in space to orbit above the distortions of Earth's atmosphere. The Hubble Space

Telescope was finally launched in 1990 and has a crystal-clear view of deep space. Named in honor of the American astronomer Edwin Hubble, it has delivered some of the sharpest pictures of celestial objects and made discoveries ranging from previously unknown moons in our solar system to distant galaxies at the edge of the universe.

Future telescope

- **What?** European Extremely Large Telescope (E-ELT)
- **Who?** European Southern Observatory
- **Where and when?** Chile, c. 2024

Currently under construction, the next generation of telescopes will be even larger and more powerful. The European Extremely Large Telescope (E-ELT) in the Atacama Desert in Chile will use a honeycomb network of mirrors whose positions are constantly adjusted by computer. With a total diameter of 129ft (39.3m), the telescope will gather 100 million times more light than a human eye. This will enable astronomers to see planets that orbit other stars and study the most distant galaxies in the universe.

